Hanjin Gong

Richard Gong · Senior Software Engineer

Financial Systems · Backend & Full-Stack Development

Tokyo, Japan · Open to relocation

Email · LinkedIn · GitHub

Profile

Software engineer with 18 years of experience across investment banking, retail banking, payments and consumer platforms. My work combines financial-domain understanding with hands-on backend, frontend and delivery engineering. At Nomura, my experience includes Java financial applications alongside JGB portfolio optimisation and risk analytics using Python and CVXPY.

Proficient in Java, Python, Go, TypeScript/JavaScript, C#, Rust and Jason/AgentSpeak, with experience spanning professional delivery, independent research and academic work. I work across requirements, implementation, integration and deployment, selecting technologies to suit the problem. My interests include financial systems, backend services, full-stack applications, quantitative development and multiagent systems.

Professional experience

Senior Engineer | Nomura Securities

Contract assignment through Michael Page Japan

Tokyo, Japan · June 2022 - Present

Platform: Japan Government Bond (JGB) Portfolio Optimisation & Risk Analytics Platform - used by rates PMs, traders, and risk officers on the Tokyo desk.

- Develop and maintain financial applications in Java at Nomura, alongside Python and TypeScript development, delivering application features and maintaining existing code.
- Built the JGB optimisation engine in Python + CVXPY: formulated risk-adjusted carry/PnL objectives under duration, DV01, issuer-concentration, liquidity-bucket, and regulatory constraints. Tuned solver (ECOS / OSQP) selection by problem shape and achieved sub-second solve times on representative books.
- Designed low-latency risk APIs in Python (Flask/FastAPI) and Node.js (NestJS) exposing optimisation, scenario PnL, and exposure-decomposition endpoints; improved tail latency through response caching, lazy curve rebuilds, and vectorised pandas hot paths.
- Re-architected the legacy monolith into Dockerised microservices on AWS (EKS + ALB), with per-service SLOs, blue/green deploys, and deployment lead time reduced from days to minutes. Introduced structured logging, Datadog dashboards, and trace propagation across the optimisation to risk to UI chain.
- Owned risk decomposition logic - positions broken down by tenor bucket, issue, and sector; key-rate DV01 and shock-scenario PnL computed incrementally to support interactive what-if flows instead of full recomputation per click.
- Built the Next.js (App Router) trader-facing front end in TypeScript: server components for heavy data fetching, streaming SSR for first paint, Zustand for interactive optimisation state. Replaced an Angular legacy path for new modules while maintaining interop during migration.
- Develop GitLab CI build and packaging jobs and Jenkins pipelines that invoke Ansible; maintain deployment playbooks and Kubernetes manifests.

Stack: Java · Python (CVXPY, NumPy, pandas, FastAPI, Flask) · Node.js (NestJS) · TypeScript / Next.js 16 · Angular · Redis · kdb+/q (read-side) · Docker · Kubernetes · AWS (EKS, ALB, S3) · GitLab CI/CD · GitLab Runner · Jenkins · Ansible · Datadog

Senior Software Engineer - Nisshin

Tokyo, Japan · Jan 2021 - Jun 2022

Platform: Talent Management System (HR SaaS)

- Built REST APIs on Java 17 + Javalin with a React / TypeScript front end for real-time HR analytics; enforced strict layering (handlers to services to repositories) to keep the codebase testable as the team grew.
- Introduced a Redis-backed caching strategy with TTL + cache-aside invalidation and rewrote N+1 MySQL queries into batched joins - cut median page-load time by ~30% and removed a recurring DB CPU spike during morning login peaks.
- Developed load and stress-testing scripts in Python and Go to evaluate application performance under concurrent workloads and support performance troubleshooting.
- Stood up GitLab CI/CD pipelines (build, test, container scan, staging deploy), mentored 5+ junior engineers on code review, branching model, and observability.

Stack: Java 17 · Javalin · Python · Go · React · TypeScript · MySQL · Redis · GitLab CI · Load & stress testing

Technical Leader - HSBC Bank

Guangzhou, China · Jan 2019 - Jan 2021

Platform: Overseas Bank Account Opening (30+ jurisdictions)

- Led architecture for a Java + Node.js microservices estate handling cross-border KYC, document capture, and account provisioning across 30+ countries with per-region compliance rules.
- Drove workflow automation (orchestration + retries + dead-letter handling) that cut manual operational overhead by ~20% and shortened average account-opening turnaround meaningfully.
- Rolled out Datadog observability across services - SLOs, latency histograms, error-budget alerting - replacing ad-hoc log grepping with actionable dashboards for on-call engineers.

Stack: Java · Spring Boot · Node.js · Oracle · MySQL · Kafka · Docker · Datadog

Technical Leader - SurLead

Manila, Philippines · Mar 2015 - Jan 2019

- Delivered FinTech and e-commerce client platforms in Java + TypeScript on Oracle / MySQL, owning architecture, delivery, and team leadership end-to-end.
- Migrated legacy deployments to Docker and split a monolith into NestJS / Express microservices - shortened release cycles by ~40% and enabled independent team ownership per service.

Stack: Java · TypeScript · NestJS · Express · Oracle · MySQL · Docker

Full-Stack Engineer - Ctrip

Shanghai, China · Jun 2013 - Feb 2015

Platform: Ctrip Wireless (travel-booking platform, millions of DAU)

- Co-designed the frontend framework (Node.js · Backbone.js · jQuery) used across booking, payment, and loyalty modules - standardised component patterns across product lines.
- Shipped user-facing features on the booking and payment critical path, serving millions of daily users with tight SLAs during peak travel seasons.

Stack: Node.js · Backbone.js · jQuery · Java (backend integrations) · MySQL

Front-End Engineer - Qunshuo

Shanghai, China · Aug 2010 - May 2013

- Built enterprise web apps on Java / Spring / Hibernate / MySQL with contemporary JS frameworks; delivered a SAP-integrated childcare module on schedule.

Core technologies

- **Languages:** Java, Python, Go, TypeScript/JavaScript, C#, Rust and Jason/AgentSpeak; additional study in Prolog.
- **Backend and APIs:** Spring Boot, Javalin, FastAPI, Flask, Node.js, NestJS, Express and Axum.
- **Frontend:** React, Next.js and Angular.
- **Programming approaches:** Object-oriented and functional programming, asynchronous and concurrent programming, Rust ownership and borrowing, logic programming and BDI agent programming.
- **Data and integration:** SQL, Oracle, MySQL, Redis, Kafka, RabbitMQ, kdb+/q, REST and gRPC.
- **Delivery:** GitLab CI/CD, Jenkins, Ansible, Docker, Kubernetes, GitHub Actions, AWS and Azure.
- **Quantitative development:** CVXPY, NumPy, pandas, portfolio optimisation, yield curves, DV01/key-rate duration and scenario PnL.

Independent Research

- **Algorithmic trading with cTrader and MetaTrader 5 (MT5):** Research automated trading strategies, signal generation, order execution, position sizing and risk management, using backtesting and forward evaluation to examine strategy behaviour.
- **Model training with Rust:** Explore Rust-based machine-learning workflows, including data preparation, model training and evaluation, with attention to computational performance and memory efficiency.
- **Trading systems with Rust:** Investigate trading-system design, including market-data processing, asynchronous execution, concurrency and order management.
- **Multiagent systems with Jason and Prolog:** Explore BDI agents, logic-based reasoning, knowledge representation, inter-agent communication and coordination for collaborative decision-making.

Education

MSc Computer Science | University of Liverpool, UK Expected 2027

- **Multiagent systems (ongoing):** Agent autonomy, reactive and proactive behaviour, BDI (beliefs, desires and intentions), deductive and practical reasoning, agent communication and coordination.
- **Logic and agent-oriented programming (ongoing):** Prolog facts, rules, queries and unification; Jason/AgentSpeak beliefs, goals, plans, events and inter-agent messaging.
- **Additional study:** Mathematical foundations for machine learning, including calculus and gradient descent; human-computer interaction and accessibility evaluation.

Associate Degree, Science & Technology | Shanghai College of Science & Technology 2007-2010

Certifications and languages

- Cisco Certified Technician (CCT) - Routing & Switching.
- Microsoft Certified Professional & Specialist (2016).
- AI, Business & the Future of Work - Lund University (2023).
- English (fluent) · Chinese (native).

Contact

Open to senior backend, full-stack and financial-software engineering opportunities, especially in trading, risk and market-data systems.

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